

Microsoft Sentinel Rogue DNS Resolver

Full Explanation With Clean Tables

This document explains how a rogue DNS resolver incident can appear in Microsoft Sentinel. It keeps the full explanation while using cleaner tables that stay inside the page margins. Long explanations are placed in the body text instead of inside table cells.

Scenario: the workstation **WKST-1042** originally used approved DNS servers **10.20.1.10** and **10.20.1.11**. The endpoint then changed to an unauthorized resolver, **203.0.113.53**. After the change, Sentinel saw DNS queries, firewall events, suspicious DNS answers, browser connections, alerts, and a final incident.

Core finding: the incident is identified through correlation, not one log line.

Key Entities

Entity Type	Value	Purpose
Endpoint hostname	WKST-1042	Workstation involved
Endpoint IP	10.20.14.42	Source of DNS traffic
User account	jsmith	User context
Approved DNS	10.20.1.10, 10.20.1.11	Expected resolvers
Unauthorized resolver	203.0.113.53	Suspicious DNS server
Suspicious returned IP	203.0.113.88	Returned for login domain
Main command tool	netsh.exe	Changed DNS settings
Main risk	DNS hijacking	Traffic redirection

Microsoft Sentinel Timeline

These are the simulated Sentinel-style events used for the investigation. The lines are wrapped to stay on the page. In a real Sentinel workspace, these would appear across tables such as ASimDns, DeviceRegistryEvents, DeviceProcessEvents, DeviceNetworkEvents, CommonSecurityLog, SecurityAlert, and SecurityIncident.

01 TimeGenerated=2026-05-08T09:14:01Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=10.20.1.10
DnsQuery=login.microsoftonline.com DnsResponseIpAddr=20.190.160.12 EventResult=Success

02 TimeGenerated=2026-05-08T09:14:07Z Table=DeviceRegistryEvents DeviceName=WKST-1042 ActionType=RegistryValueSet
RegistryKey=HKLM\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\{NIC-GUID} RegistryValueName=NameServer
PreviousValue="10.20.1.10,10.20.1.11" NewValue="203.0.113.53" InitiatingProcessFileName=netsh.exe AccountName=jsmith

03 TimeGenerated=2026-05-08T09:14:07Z Table=DeviceProcessEvents DeviceName=WKST-1042 ActionType=ProcessCreated FileName=netsh.exe
ProcessCommandLine="netsh interface ip set dns name=Ethernet static 203.0.113.53" InitiatingProcessFileName=cmd.exe AccountName=jsmith

04 TimeGenerated=2026-05-08T09:14:10Z Table=DeviceNetworkEvents DeviceName=WKST-1042 ActionType=ConnectionSuccess LocalIP=10.20.14.42
RemoteIP=203.0.113.53 RemotePort=53 Protocol=Udp InitiatingProcessFileName=svchost.exe AccountName=jsmith

05 TimeGenerated=2026-05-08T09:14:10Z Table=CommonSecurityLog DeviceVendor=Firewall DeviceAction=Allow SourceIP=10.20.14.42
DestinationIP=203.0.113.53 DestinationPort=53 Protocol=UDP ApplicationProtocol=DNS RuleName=Allow_Outbound_DNS

06 TimeGenerated=2026-05-08T09:14:11Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=203.0.113.53 DnsQuery=login.microsoftonline.com DnsResponseIpAddr=203.0.113.88 EventResult=Success

07 TimeGenerated=2026-05-08T09:14:12Z Table=SecurityAlert AlertName="DNS Query To Unauthorized Resolver" ProviderName="Microsoft Sentinel" AlertSeverity=Medium CompromisedEntity=WKST-1042 Entities="10.20.14.42,203.0.113.53"

08 TimeGenerated=2026-05-08T09:14:15Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=203.0.113.53 DnsQuery=vpn.company-portal.com DnsResponseIpAddr=203.0.113.89 EventResult=Success

09 TimeGenerated=2026-05-08T09:14:18Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=203.0.113.53 DnsQuery=autodiscover.company.com DnsResponseIpAddr=203.0.113.90 EventResult=Success

10 TimeGenerated=2026-05-08T09:14:21Z Table=DeviceNetworkEvents DeviceName=WKST-1042 ActionType=ConnectionSuccess LocalIP=10.20.14.42 RemoteIP=203.0.113.88 RemotePort=443 Protocol=Tcp RemoteUrl=login.microsoftonline.com InitiatingProcessFileName=chrome.exe AccountName=jsmith

11 TimeGenerated=2026-05-08T09:14:22Z Table=SecurityAlert AlertName="Suspicious DNS Answer For High-Value Domain" ProviderName="Microsoft Sentinel" AlertSeverity=High CompromisedEntity=WKST-1042 DnsQuery=login.microsoftonline.com ReturnedIP=203.0.113.88 ExpectedResolvers="10.20.1.10,10.20.1.11"

12 TimeGenerated=2026-05-08T09:14:28Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=10.20.1.10 DnsQuery=login.microsoftonline.com EventResult=Failure EventMessage="No recent query from endpoint after DNS configuration change"

13 TimeGenerated=2026-05-08T09:14:35Z Table=CommonSecurityLog DeviceVendor=Firewall DeviceAction=Allow SourceIP=10.20.14.42 DestinationIP=203.0.113.53 DestinationPort=53 Protocol=TCP ApplicationProtocol=DNS RuleName=Allow_Outbound_DNS

14 TimeGenerated=2026-05-08T09:14:39Z Table=ASimDns EventType=Query SrcIpAddr=10.20.14.42 SrcHostname=WKST-1042 DnsServerIpAddr=203.0.113.53 DnsQuery=download.windowsupdate.com DnsResponseIpAddr=203.0.113.91 EventResult=Success

15 TimeGenerated=2026-05-08T09:14:43Z Table=SecurityAlert AlertName="Endpoint DNS Server Changed To External Resolver" ProviderName="Microsoft Sentinel" AlertSeverity=High CompromisedEntity=WKST-1042 AccountName=jsmith OldDnsServers="10.20.1.10,10.20.1.11" NewDnsServer="203.0.113.53"

16 TimeGenerated=2026-05-08T09:14:52Z Table=SecurityAlert AlertName="Possible Rogue DNS Resolver In Use" ProviderName="Microsoft Sentinel" AlertSeverity=Critical CompromisedEntity=WKST-1042 SourceIP=10.20.14.42 UnauthorizedDnsServer=203.0.113.53 ApprovedDnsServers="10.20.1.10,10.20.1.11" EventCount=6 TimeWindow=45s

17 TimeGenerated=2026-05-08T09:15:03Z Table=DeviceProcessEvents DeviceName=WKST-1042 ActionType=ProcessCreated FileName=ipconfig.exe ProcessCommandLine="ipconfig /flushdns" InitiatingProcessFileName=cmd.exe AccountName=jsmith

18 TimeGenerated=2026-05-08T09:15:09Z Table=SecurityAlert AlertName="DNS Configuration Modified By Command Line" ProviderName="Microsoft Sentinel" AlertSeverity=High CompromisedEntity=WKST-1042 ProcessName=netsh.exe ProcessCommandLine="netsh interface ip set dns name=Ethernet static 203.0.113.53"

19 TimeGenerated=2026-05-08T09:15:22Z Table=CommonSecurityLog DeviceVendor=Firewall DeviceAction=Block SourceIP=10.20.14.42 DestinationIP=203.0.113.53 DestinationPort=53 Protocol=UDP ApplicationProtocol=DNS RuleName=Block_Unauthorized_DNS

20 TimeGenerated=2026-05-08T09:15:31Z Table=SecurityIncident IncidentName="Rogue DNS Resolver - WKST-1042" Severity=High Status=New Owner=SOC Entities="WKST-1042,10.20.14.42,203.0.113.53,jsmith" Classification=Undetermined

Clean Line-by-Line Identification Table

This table keeps the cells short so it remains readable. The detailed explanation follows immediately after the table.

Line	Sentinel Table	Key Evidence	What It Proves
01	ASimDns	Approved DNS query	Normal baseline
02	DeviceRegistryEvents	NameServer changed	DNS config changed
03	DeviceProcessEvents	netsh.exe set DNS	Command-line method
04	DeviceNetworkEvents	UDP/53 to 203.0.113.53	Active rogue DNS use
05	CommonSecurityLog	Firewall allowed DNS	Path allowed
06	ASimDns	Login domain returned 203.0.113.88	Suspicious DNS answer

Line	Sentinel Table	Key Evidence	What It Proves
07	SecurityAlert	Unauthorized resolver alert	First detection
08	ASimDns	VPN domain via rogue DNS	Sensitive domain affected
09	ASimDns	Autodiscover via rogue DNS	Mail config affected
10	DeviceNetworkEvents	Chrome to returned IP:443	DNS influenced traffic
11	SecurityAlert	High-value DNS answer alert	Severity escalated
12	ASimDns	No normal internal query	DNS bypass evidence
13	CommonSecurityLog	TCP/53 to rogue DNS	Continued use
14	ASimDns	Windows Update via rogue DNS	Update domain affected
15	SecurityAlert	DNS changed alert	High-confidence change
16	SecurityAlert	Critical rogue DNS alert	Correlation alert
17	DeviceProcessEvents	ipconfig /flushdns	Cache cleared
18	SecurityAlert	Command-line DNS alert	netsh confirmed
19	CommonSecurityLog	Firewall blocked rogue DNS	Containment
20	SecurityIncident	Incident created	SOC case opened

Detailed Line-by-Line Explanation

01 - Baseline DNS Query

The first event shows WKST-1042 resolving login.microsoftonline.com through 10.20.1.10. This matters because 10.20.1.10 is an approved corporate DNS server. This line establishes what normal looked like before the change.

02 - DNS Registry Value Changed

DeviceRegistryEvents shows the Windows NameServer value changing from 10.20.1.10,10.20.1.11 to 203.0.113.53. This is one of the strongest indicators because it proves the endpoint DNS configuration changed to an external resolver.

03 - netsh.exe Made the Change

DeviceProcessEvents shows netsh.exe running a command to set Ethernet DNS to 203.0.113.53. This explains how the change likely happened and ties the configuration change to a command-line action launched from cmd.exe.

04 - Endpoint Communicates With Rogue Resolver

DeviceNetworkEvents shows WKST-1042 connecting to 203.0.113.53 on UDP port 53. This proves the endpoint is actively using the unauthorized DNS server, not merely configured to use it.

05 - Firewall Allows the DNS Flow

CommonSecurityLog confirms the firewall allowed 10.20.14.42 to reach 203.0.113.53 on port 53. This validates the network path and shows why the DNS bypass succeeded at first.

06 - High-Value Domain Resolved Through Rogue DNS

ASimDns shows login.microsoftonline.com queried through 203.0.113.53 and returning 203.0.113.88. This is high risk because Microsoft login traffic is sensitive, and a rogue resolver can redirect users to fake login infrastructure.

07 - First Sentinel Alert

SecurityAlert shows DNS Query To Unauthorized Resolver. This is the first detection-level confirmation that Sentinel saw DNS traffic to a resolver outside the approved list.

08 - VPN Domain Uses Rogue DNS

The endpoint resolves vpn.company-portal.com through 203.0.113.53. VPN domains are sensitive because redirection could lead to credential harvesting or fake portal infrastructure.

09 - Autodiscover Uses Rogue DNS

The endpoint resolves autodiscover.company.com through the unauthorized resolver. Autodiscover is tied to mail and client configuration, so manipulation here can affect authentication or email-related workflows.

10 - Browser Connects to Returned IP

DeviceNetworkEvents shows chrome.exe connecting to 203.0.113.88 over TCP 443 with RemoteUrl=login.microsoftonline.com. This is a major pivot because the browser connects to an IP returned by the rogue resolver.

11 - Suspicious DNS Answer Alert

Sentinel raises Suspicious DNS Answer For High-Value Domain. The suspicious returned IP for a sensitive domain increases severity because the issue is no longer only unauthorized DNS usage; it may be redirection.

12 - Internal DNS Visibility Drops

ASimDns shows no recent normal query from the endpoint after the DNS configuration change. This supports DNS bypass because internal DNS no longer has normal visibility into that host.

13 - DNS Over TCP Also Observed

CommonSecurityLog shows TCP port 53 traffic to 203.0.113.53. DNS normally uses UDP, but TCP 53 can be legitimate for fallback or larger responses. Seeing both UDP and TCP strengthens the evidence of active rogue resolver use.

14 - Windows Update Domain Affected

ASimDns shows download.windowsupdate.com resolving through 203.0.113.53 to 203.0.113.91. This is another high-value domain because update-related redirection can support fake updates or malware delivery.

15 - DNS Change Alert

SecurityAlert states Endpoint DNS Server Changed To External Resolver. This is high-confidence because the alert directly identifies the endpoint configuration change.

16 - Main Correlation Alert

SecurityAlert shows Possible Rogue DNS Resolver In Use with Critical severity, EventCount=6, and TimeWindow=45s. This is the main correlation point where Sentinel combines repeated unauthorized DNS activity into a higher-confidence incident signal.

17 - DNS Cache Flushed

DeviceProcessEvents shows ipconfig.exe running ipconfig /flushdns. Alone this is normal, but after a DNS change it is suspicious because flushing DNS forces the endpoint to discard old cached answers and use the new resolver immediately.

18 - Command-Line DNS Modification Alert

SecurityAlert references DNS Configuration Modified By Command Line and the netsh.exe command. This ties the DNS change to command-line behavior, which can indicate malware, a script, or hands-on-keyboard activity.

19 - Firewall Containment

CommonSecurityLog shows the firewall blocking DNS traffic to 203.0.113.53. This shows containment because the organization has started blocking the unauthorized resolver.

20 - Sentinel Incident Created

SecurityIncident shows Rogue DNS Resolver - WKST-1042. Sentinel packages the host, user, endpoint IP, and rogue resolver into a SOC case for investigation.

Detection Chain

Step	Evidence	Meaning
1	WKST-1042 -> 10.20.1.10	Approved DNS baseline
2	NameServer -> 203.0.113.53	DNS changed
3	netsh.exe set DNS	Command-line change
4	10.20.14.42 -> 203.0.113.53:53	Rogue resolver used
5	High-value domains resolved	Sensitive traffic risk
6	chrome.exe -> 203.0.113.88:443	Browser followed result
7	SecurityIncident created	SOC case opened

Why This Is Identified as Rogue DNS

This is not identified from one log line alone. The confidence comes from correlation. Sentinel first sees the endpoint using approved DNS. Then it sees a Windows DNS registry value change. Next, it sees netsh.exe setting the external DNS server. After that, endpoint, DNS, and firewall logs show traffic to 203.0.113.53 on port 53.

The activity becomes more serious because high-value domains resolve through the unauthorized resolver. The endpoint then connects to an IP returned by the rogue resolver. This creates a logical chain from configuration change to DNS bypass to possible redirection.

Most Important Evidence

Priority	Line	Evidence	Reason
1	02	DNS registry changed	Proves configuration change
2	03	netsh.exe command	Shows likely method
3	06	Login domain via rogue DNS	Sensitive query affected
4	10	Chrome to returned IP	Traffic followed answer
5	16	Critical Sentinel alert	Correlation confirmed

What This Could Indicate

Possibility	Why It Fits
Malware	Can change DNS using netsh or registry edits
Manual attacker action	Command-line DNS change observed
Malicious script	cmd.exe launched netsh.exe
VPN/proxy issue	Some tools alter DNS settings
DHCP problem	Possible, but local netsh is stronger evidence
DNS hijacking	Legitimate domains may be redirected
Phishing redirection	Login domains are high-value targets

What a SOC Analyst Should Investigate Next

Investigation Area	What to Check
Process lineage	Parent process of netsh.exe
User context	Whether jsmith normally changes DNS
Endpoint timeline	Downloads, scripts, remote tools, prompts
DNS scope	Other hosts using 203.0.113.53

Investigation Area	What to Check
Firewall scope	UDP/TCP 53 to unauthorized resolvers
DoH bypass	DNS-over-HTTPS over port 443
Browser/proxy logs	Login pages after DNS change
Credential risk	Credentials entered after redirection
Containment	Restore DNS and block rogue resolver

Clean Interview Explanation

This Microsoft Sentinel timeline identifies a rogue DNS resolver because WKST-1042 originally used approved DNS servers, but then its DNS configuration changed to 203.0.113.53. The strongest proof is in DeviceRegistryEvents, where the Windows NameServer value changes from 10.20.1.10,10.20.1.11 to 203.0.113.53. DeviceProcessEvents then shows netsh.exe being used from cmd.exe to set the DNS server through the command line.

After the DNS change, DeviceNetworkEvents, CommonSecurityLog, and ASimDns show the endpoint sending DNS traffic to 203.0.113.53 on port 53. The rogue resolver returns suspicious IP addresses for high-value domains like login.microsoftonline.com, and the endpoint later connects to one of those returned IPs over HTTPS using Chrome. Sentinel then correlates the DNS setting change, unauthorized resolver usage, suspicious DNS answers, firewall traffic, and command-line modification into a high or critical incident.

I would investigate whether this was caused by malware, a script, VPN software, DHCP misconfiguration, or manual user/admin activity. I would restore approved DNS settings, block outbound DNS except to approved resolvers, hunt for other endpoints using 203.0.113.53, review browser and proxy activity after the DNS change, and check whether credentials were entered into any redirected site.